

Akash Purushotham

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PROFESSIONAL SUMMARY

Cloud-native Java developer with 3+ years of experience building and optimizing large-scale applications, REST APIs, and multi-threaded systems. Strong expertise in software development, containerization, and migrating legacy applications to modern cloud architectures with a focus on performance optimization and scalability.

TECHNICAL SKILLS

Languages: Java, C++, Python, JavaScript, SQL, Bash
Data: Apache Spark, Trino, Hadoop, Kafka, Alluxio
Backend: Node.js, Spring Boot, Maven, JUnit, MVC-Arch
Core: Data Structures, Algorithms, Design Patterns
AI: LLM, MCP, RAG, Copilot, SDV, Agentic AI

DevOps: Kubernetes, Docker, Helm, CI/CD, GitHub Actions
Cloud: AWS, Azure, GCS, S3, EMR, EKS, Linux
Database: MongoDB, PostgreSQL, Redis, Clickhouse
Others: Git, Jenkins, Keycloak, Terraform, Distributed Systems
AI Ops: Prometheus, Fluentbit, ELK, JMX, Grafana

PROFESSIONAL EXPERIENCE

- Mobileum Technologies** Bengaluru, India
Sept 2022 - Present
 - Software Engineer I → Software Engineer II
 - Built a PySpark-based synthetic data generation platform capable of producing realistic large-scale time-series datasets for analytics, testing, and AI/ML workloads, leveraging distributed Spark processing, configurable data modeling, and cloud-native storage integration.
 - Led cloud migration of multiple on-premise data platforms to Kubernetes, standardizing deployment, improving scalability, and reducing operational complexity by 30%.
 - Extended the Trino distributed query engine with custom Java plugins and 10+ analytical functions, including in-query data enrichment and adaptive cardinality estimation using HyperLogLog, eliminating 50% of expensive JOIN operations, reducing memory consumption by up to 98%, and improving query performance by 90%.
 - Modernized a Java/Scala-based distributed analytics and data warehousing platform built on Apache Spark, Hadoop, HBase, Presto, Alluxio, and Zookeeper by migrating from Java 8 to Java 17, containerizing workloads on Kubernetes, and enabling multi-cloud storage integration (AWS S3, Azure Blob Storage, GCS), reducing infrastructure costs by 20% and simplifying deployments.
 - Designed and developed a Java 17/Quarkus-based Spark Job Manager platform to orchestrate distributed Spark workloads across Kubernetes and local environments, providing automated job submission, monitoring, lifecycle management, and seamless integration with AWS S3, Azure Blob Storage, and Google Cloud Storage.
 - Led the migration of large-scale analytics workloads from Apache Hive to ClickHouse by developing a custom connector, production-ready Docker images, and Kubernetes Helm charts, delivering 3x faster query performance, up to 40% lower infrastructure costs, and improved cloud-native scalability.
 - Architected and developed a Spring Boot-based Revenue Share Calculator (RSC) analytics microservice leveraging ClickHouse and PostgreSQL to process terabyte-scale roaming and subscriber data, delivering low-latency distributed querying, stream-based exports, and 10–100x performance improvements.
 - Designed and developed a production-grade Java 21 data ingestion and synchronization platform (DBSync) integrating Oracle, Couchbase, Kafka, and ClickHouse to synchronize 35 telecom reference datasets across distributed analytics clusters using atomic table swaps, ensuring zero-downtime refreshes, data consistency, and fault-tolerant operation on Kubernetes.
 - Developed a containerized Node.js/Playwright reporting platform that automatically generated and distributed PDF exports of analytics dashboards, reducing manual reporting effort by 85%.
 - Built a C++ CDR logging framework for telecom applications and resolved critical memory-management issues using Valgrind, reducing production crashes by 30% and improving operational observability.

PERSONAL PROJECTS

- Tech Legends** github.com/itsakashp/Tech-Legends
 - Full Stack Web Application - Node.js, Express.js, MongoDB, EJS, Bootstrap 2022
 - Created a platform showcasing influential technology pioneers with user authentication and media storage

EDUCATION

- Jain University** Bengaluru, India
 - B.Tech in Computer Science and Engineering - GPA: 8.5/10.0